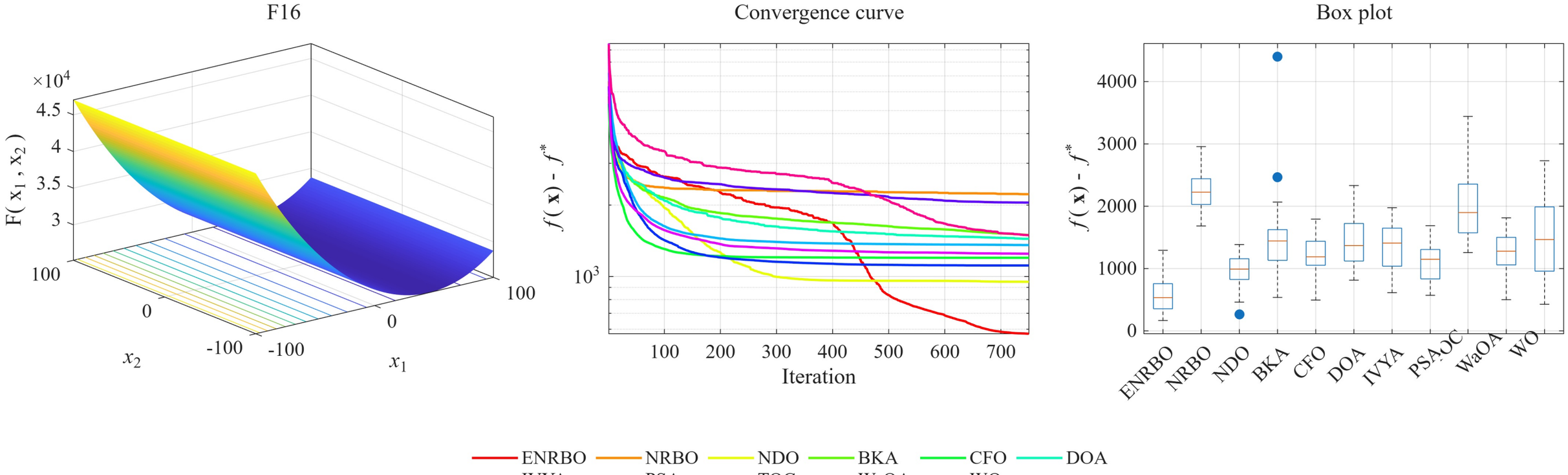

F16
×10^4
4.5
4
3.5
3
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
f (x) - f*
10^3
100
200
300
400
500
600
700
Iteration
Box plot
f (x) - f*
4000
3000
2000
1000
0
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO